

Stress and friction evolution in the bolt during and after assembly[☆]

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ABSTRACT

The stress distribution within threaded connections exhibits significant complexity attributable to the intricate interfacial friction between mating surfaces, which critically influences the coupled relationship among tightening torque, rotational angle, and axial preload. This study establishes a comprehensive assembly analytical framework incorporating three critical factors: thread contact deformation, screw body distortion, and frictional distribution evolution during and after assembly processes. Through this model, three distinct deformation patterns and corresponding stress distribution characteristics in bolted joints have been systematically identified. Subsequent numerical simulations quantitatively reveal the generation mechanisms of these stress patterns under varying operational conditions. Particularly, parametric analysis demonstrates that reduced stiffness of clamped components amplifies both rotational displacement and axial deformation of fasteners. The proposed methodology further elucidates the mechanism underlying post-assembly preload relaxation phenomena. Experimental validation through quasi-static testing confirms measurement consistency across torque-preload-angle parameters, with theoretical predictions showing excellent consistency with experimental data. This analytical advancement enhances fundamental understanding of bolted joint mechanics while providing theoretical guidance for engineering applications in precision assembly and structural safety assessment.

1. Introduction

Threaded connections are simple in structure and easy to disassemble, making them a common connection in engineering applications. They are widely used in various industries such as aerospace, mechanical and civil engineering [1]. During the assembly and vibration environment of a bolt, whether a threaded connection will fail or not is directly related to the magnitude of the contact force between the threads. Therefore, in order to calibrate the connection reliability, it is necessary to study the stress and load distribution in bolts [2]. However, it is difficult to develop an accurate mathematical model of the bolt assembly process due to the complexity of inter-thread contact friction and deformation [3].

During the assembly process of threaded connections, it is critical to figure out the contact characteristics between the bolt head and the connecting part, as well as the contact characteristics between the threads. Motosh developed a friction model to derive an accurate torque-axial force relationship in bolt, i.e., the tightening torque is equal to the sum of the thread contact torque and the underhead bearing friction torque [4]. After calculation, it can be found that the preload force is positively proportional to the torque, and the proportionality coefficient is related to the equivalent friction radius

and the friction coefficient. Later on, Nassar et al. proposed four different underhead pressure distribution models and five different pressure distribution models between threads [5,6]. According to their studies, the new equivalent friction radius can be obtained to replace the mean friction radius used in Motosh's model. Shi et al. transformed the contact pressure distribution into normal contact load distribution to study the contact pressure distribution and its mathematical representation of single-bolted joint [7]. In terms of research on thread contact deformation, Den Hartog viewed the screw threads as cantilever beams subjected to bending and shear, and the contact force between threads was assumed to be a normal concentrated force acting at the midpoint of the cantilever beam [8]. Sopwith expanded the above method by considering radial compression of threads and radial displacement of nuts and bolts due to Poisson's ratio effect, and obtained a widely accepted load distribution model [9]. Yamamoto proposed a more complete analysis of the deformation of the screw thread cross-section [10]. He categorized the elastic deformation of screw threads into the following five types: bending deformation δ_1 , shearing deformation δ_2 , inclination deformation of the thread root δ_3 , shearing deformation of the thread root δ_4 , and deformation due to radial expansion (nut) and radial shrinkage (bolt) δ_5 . Recently, some scholars have conducted

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research based on Yamamoto's model. Zhang et al. developed a computational method for threaded connection stiffness and verified it through tensile tests and finite element (FE) analysis [11]. Lu et al. obtained a modified load distribution model by considering the effect of friction coefficient, and verified the theoretical results using the same method as Zhang's study [12,13]. The above theoretical study only focuses on the axial load distribution of threaded connections, without considering torsion. However, the torsion effect of bolts is unavoidable in practical engineering. Shi established a comprehensive model which takes tension-torsion coupling into consideration for thread connection for the first time [14].

In order to obtain the stress distribution in bolts more intuitively, a variety of experiments were designed and conducted. Goodier measured the radial and axial deformation of the nut using a tensiometer to obtain the axial load distribution [15]. The load was found to be small at the free end or in the middle section, but concentrated near the loaded end. At higher loads this concentration was not significant. Subsequently, with the development of photoelasticity technology, some scholars have used three-dimensional(3D) photoelasticity experiments to study the stress distribution in bolts [16–18]. They found that the experimental results were in general agreement with Sopwith's theoretical results except for the loaded face of the nut. This discrepancy may be due to the fact that nut threads near the loaded face were not fully formed and therefore exhibit a lower stiffness and load bearing capacity than fully formed threads. Some scholars have used other means to measure bolt stress distribution, including copper electroplating method [19,20], nut deformation measurement [21] and bolt deformation measurement [22]. These methods require a large number of discrete measurement points and measurement accuracy cannot be guaranteed. In recent years, ultrasonic axial stress measurement methods have been used for the bolt stress detection [23]. These methods are suitable for the detection of axial stress in an assembly or a tightened bolt under elastic deformation. Our research team proposed a means of real-time stress measurement using optical fiber OFDR technology, which can achieve accurate measurement of bolt axial stress [24].

With the development of computer technology, FE analysis has also become an important means of studying stress distribution in bolts [25–32]. Due to the complex contact interface of the threaded connections, certain assumptions and simplifications were made in the FE models. Initially, some scholars approximated the threaded connection problem to be axially symmetric. Maruyama used an axisymmetric FE model to investigate the stresses in the nut-bolt connection, ignoring the effect of friction and considering only the axial forces [25]. Fukuoka et al. developed an axisymmetric FE model that takes into account axial loads as well as torsional loads generated by friction [33,34]. Honglin Xu et al. studied the stress distribution of casing threaded joints according to the Yamamoto method, and established a two-dimensional(2D) FE model [19]. In order to accurately study the thread contact stress, 3D FE models were established. Brutti developed a method to calculate the state of stress at the root of the thread taking into account the effects of the washer, and established a FE model for ISO M27 to check the accuracy of the proposed method. Shikun Lu compared the stress distribution of Yamamoto model with the 3D FE model, and found the load and stress distribution results of the two methods are basically the same [12].

In the previous theoretical models, the axial stress distribution in bolts was basically obtained based on the studies of Sopwith [9] and Yamamoto [10]. Although these studies can obtain axial stress distribution in bolts in general agreement with experimental and FE analysis, the role of torsion is not taken into account. Some scholars have considered the effect of torsion, but not the effect of tightening torque in actual bolt assembly [14,33]. There are two steps involved in the bolt assembly process, i.e., the addition and removal of the tightening torque. During the addition of tightening torque, the threaded connection can reach a state of equilibrium. If the tightening torque is removed, the stress distribution on the contact surfaces will change and

a new equilibrium will be achieved. Therefore, the aim of this study is to develop a new theoretical model of load distribution which takes into account the coupling of tensile and torsional deformations during the bolt assembly process.

In this article, we will study the torsional and tensile deformation of bolts during the torque addition process. Compared to the deformation of the bolt, the deformation of the nut is negligible [12]. Therefore, only the deformation of the nut threads is considered. By establishing the differential equilibrium equations for torsion and tension of bolts, and combining them with boundary conditions, the deformation and stress distribution of bolts can be obtained. Consistent with the Yamamoto model, the thread is considered as a cantilever beam [10]. The friction between the contact surfaces satisfies Coulomb's friction law. The stress redistribution modeling after torque removal is similar to the above approach. In this case, the question is whether the frictional moment under the bolt head is sufficient to counteract the thread contact torque. If the underhead friction coefficient is sufficiently large, the stress distribution in bolts will not change. Otherwise, the direction of the partial friction on the thread contact surfaces will change to achieve a new equilibrium.

The rest of this paper is organized as follows. In Section 2, the relationship between the tightening torque T and the preload force F is deduced. In Section 3, stress distribution of bolt during the tightening process is studied. In Section 4, stress redistribution model after removal of tightening torque is established. In Sections 5 and 6, relevant numerical simulations and experiments are carried out and compared with the proposed model. Finally, the conclusions are summarized in Section 7.

2. Assembly torque model

The purpose of this section is to establish the relationship between tightening torque T and preload force F . Our research team has conducted a relatively mature analysis of this assembly torque model [35]. In order to facilitate the understanding of the subsequent stress analysis, it is described in detail in this section. A brief analysis shows that during the bolt assembly process, as depicted in Fig. 1(a), the tightening torque T is equal to the sum of the underhead contact torque T_t and the thread contact torque T_b . In the following, T_t and T_b are analyzed separately.

2.1. Underhead contact torque T_t

The underhead contact torque is generated by the underhead friction during the assembly process, and the magnitude of the friction is related to the distribution of normal compressive stress. Therefore, before analyzing the underhead contact torque, the stress distribution on the contact surface need to be determined. The following reasonable assumptions are made for the contact stress:

- i. The bolt head and the support surface are assumed to be rigid bodies.
- ii. The contact stress between the bolt head and the support surface are uniformly distributed.

On the contact surface, as shown in Fig. 1(b), the moment generated by friction can be calculated as

$$T_t = \iint \mu_t \sigma r dA,$$

where normal stress $\sigma = F/A_h$, A_h is the contact area of the bolt head, μ_t is the underhead friction coefficient. So the above equation can be formulated as

$$T_t = \mu_t F r_t, \quad (1)$$

where $r_t = \iint r dA/A_h$. For the hex head cap bolt shown in Fig. 1(b),

$$r_t = \frac{4}{3\sqrt{3}a_h^2 - 2\pi r_i^2} \left[\frac{3\sqrt{3}}{4} \left(\frac{1}{3} + \frac{1}{4} \ln 3 \right) a_h^3 - \frac{\pi}{3} r_i^3 \right],$$

where r_i is half the nominal radius of the bolt, and a_h is the side length of the hex head cap.

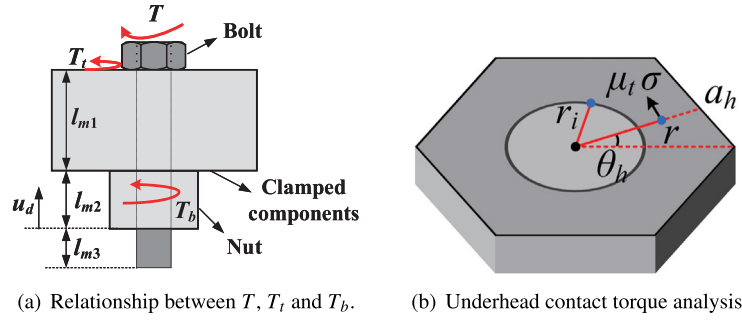


Fig. 1. Assembly torque analysis.

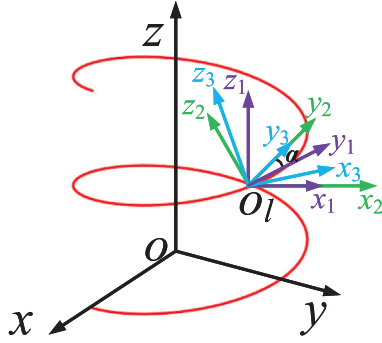


Fig. 2. Coordinate systems on the helix.

2.2. Thread contact torque T_b

In order to calculate the thread contact torque, it is necessary to obtain the contact force. According to Yamamoto method [10], the bolt thread can be simplified to a circular helix of radius a (half of the effective thread diameter), assuming that the thread contact force acts on the helix. The helix equation can be expressed as

$$x = a \cos \theta, \quad y = a \sin \theta, \quad z = b \theta,$$

where θ denotes the angular coordinate of a point on the helix, $b = p/(2\pi)$, p is the pitch of thread.

For the purpose of establishing the relationship between axial and tangential forces in a bolt, the global and local coordinate systems are established separately on the helix as follows. As shown in Fig. 2, $e(x, y, z)$ is the global coordinate system with the origin at the bottom center of the bolt, z -axis along the axial direction of the bolt, and x -axis pointing to the start of the helix. For any point O_l on the helix, there are three local coordinates. The direction of $e_1(x_1, y_1, z_1)$ can be obtained by rotating e counterclockwise about the z -axis by θ_l (the coordinates of the helix angle corresponding to point O_l). $e_2(x_2, y_2, z_2)$ is obtained by rotating e_1 counterclockwise about the x_1 -axis by α (thread lead angle). $e_3(x_3, y_3, z_3)$ is obtained by rotating e_2 counterclockwise about the y_2 -axis by β (half of thread included angle).

The transformations between these local coordinate systems are as follows.

$$e_1^T = e^T A_1, \quad e_2^T = e_1^T A_2, \quad e_3^T = e_2^T A_3,$$

where A_1, A_2, A_3 are the coordinate transformation matrices.

$$A_1 = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha & -\sin \alpha \\ 0 & \sin \alpha & \cos \alpha \end{bmatrix},$$

$$A_3 = \begin{bmatrix} \cos \beta & 0 & \sin \beta \\ 0 & 1 & 0 \\ -\sin \beta & 0 & \cos \beta \end{bmatrix}.$$

The contact forces acting on the thread surface are assumed to be the normal force f and the friction force μf , where μ is the coefficient of friction between the threads. Then the thread contact force can be expressed in e_3 coordinate system as $(0, \mu f, -f)$. According to the coordinate transformation, in e_1 coordinate system, the projections of the contact forces on the z_1 and y_1 coordinate axes are

$$f_{z1} = (\mu \sin \alpha - \cos \alpha \cos \beta) f = -\sqrt{\mu^2 + \cos^2 \beta} \cos(\alpha + \xi) f,$$

$$f_{y1} = (\mu \cos \alpha + \sin \alpha \cos \beta) f = \sqrt{\mu^2 + \cos^2 \beta} \sin(\alpha + \xi) f,$$

where $\xi = \arctan(\mu / \cos \beta)$.

The thread contact torque satisfies $T_b = \int_0^s f_{y1} ds$, the preload force satisfies $F = \int_0^s f_{z1} ds$, s is the arc length coordinate of a point on the helix. Thus, we can obtain the relationship between thread contact torque T_b and preload force F .

$$T_b = F a \tan(\xi + \alpha). \quad (2)$$

Then the tightening torque can be expressed as

$$T = T_t + T_b = k_{FT} F, \quad (3)$$

where $k_{FT} = \mu_t r_t + a \tan(\xi + \alpha)$. Ignoring the value of $\tan \xi \tan \alpha$, this result is the same as that of Motosh [4].

3. Stress distribution in the bolt during the torque addition process

The complete bolt assembly process consists of both tightening torque addition and removal. This section will present a detailed analysis for the stress distribution in the bolt during the tightening torque addition process. We have previously established a stress distribution model [35], but the influence of the stiffness of the clamped components is not considered, and the consistency between the established model and the experiments was not as expected. Therefore, in this paper, the stiffness of the clamped components is considered for modeling. First, according to the Yamamoto model, the thread contact stiffness can be obtained. Then the tensile and torsional equilibrium equations of the bolt can be obtained from the element force analysis. Finally, through the deformation evolution of the thread contact points during the tightening process, the supplementary deformation conditions can be summarized, and the equilibrium equation can be solved by combining with the boundary conditions.

3.1. Thread contact stiffness

As shown in Fig. 3, based on the Yamamoto model, the thread contact deformation is categorized into five types under the action of the normal contact force f [10]: bending deformation δ_1 , shearing deformation δ_2 , inclination deformation of the thread root δ_3 , shearing deformation of the thread root δ_4 , and deformation due to radial expansion (nut) and radial shrinkage (bolt) δ_5 .

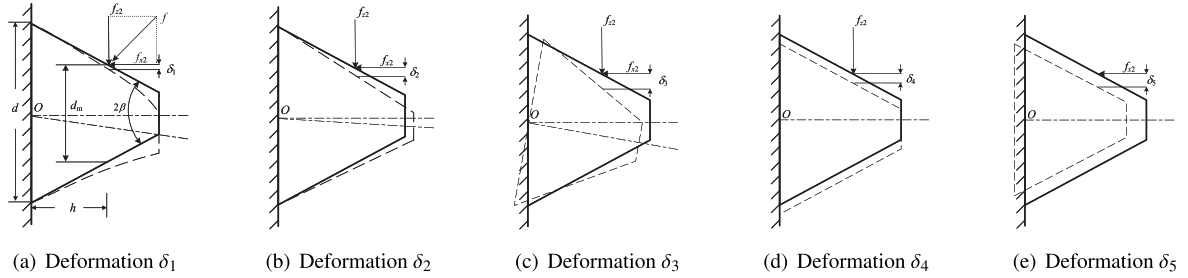


Fig. 3. Thread contact deformation.

The relationship between these five kinds of deformation and the normal contact force f are as follows.

$$\begin{aligned}\delta_{1i} &= \frac{3f \cos \beta (1 - \nu_i^2)}{4E_i} \left\{ \left[1 - \left(2 - \frac{d_m}{d} \right)^2 + 2 \ln \frac{d}{d_m} \right] \cot^3 \beta \right. \\ &\quad \left. - 4 \left(\frac{h}{d} \right)^2 \tan \beta \right\}, \\ \delta_{2i} &= \frac{6f \cos \beta \cot \beta (1 + \nu_i)}{5E_i} \ln \frac{d}{d_m}, \\ \delta_{3i} &= \frac{12f \cos \beta (1 - \nu_i^2)h}{\pi E_i d^2} \left(h - \frac{d_m}{2} \tan \beta \right), \\ \delta_{4i} &= \frac{2f \cos \beta (1 - \nu_i^2)}{\pi E_i} \left[\frac{p}{d} \ln \frac{p + d/2}{p - d/2} + \frac{1}{2} \ln \left(4 \frac{p^2}{d^2} - 1 \right) \right], \\ \delta_{5i} &= \begin{cases} \frac{f d_p \cos \beta \tan^2 \beta (1 - \nu_i)}{2p E_i}, & \text{for } i = s \\ \frac{f d_p \cos \beta \tan^2 \beta}{2p E_i} \left(\frac{d_0^2 + d_p^2}{d_0^2 - d_p^2} + \nu_i \right), & \text{for } i = m \end{cases}\end{aligned}$$

where $i = s$ or m denotes the bolt or nut, respectively. E is Young's modulus, ν is Poisson's ratio, $d_p = 2a$ is the effective diameter of the thread, d_0 is the outer diameter of the nut, and p is the pitch of thread. h is the thread pitch line height, d is the thread root width, d_m is the thread width at the pitch line. According to the Yamamoto model, the value of h, d_m, d can be obtained.

$$h = \begin{cases} 0.289p, & \text{for the bolt thread} \\ 0.325p, & \text{for the nut thread} \end{cases} \quad d = \begin{cases} 0.833p, & \text{for the bolt thread} \\ 0.875p, & \text{for the nut thread} \end{cases}$$

$$d_m = 0.5p.$$

The thread contact stiffness of the bolt and nut can then be expressed as

$$k_i = f \cos \beta / \delta_i, \quad (4)$$

where $\delta_i = \delta_{1i} + \delta_{2i} + \delta_{3i} + \delta_{4i} + \delta_{5i}$.

3.2. Tensile and torsional equilibrium equations

After obtaining the thread contact stiffness, we can proceed to establish the tensile and torsional equilibrium equations of the bolt.

As shown in Fig. 4, assume that the length of the clamped components is l_{m1} , the length of thread engagement (effective height of nut) is l_{m2} , and the length of the screw is l . The stiffness of the clamped components is k_c and the compressive deformation of the clamped components $\delta_c = P/k_c$, where P denotes the preload. The origin of the global coordinate system is located at the bottom center of the nut. The contact points in the bolt and nut are represented by the axial coordinates z_s and z_m , respectively. During tightening of the bolt, the preload is exactly 0 when the bolt head is exactly in contact with the clamped components, and further tightening will cause the preload to increase gradually. The position of the bolt at this time is set to its initial position.

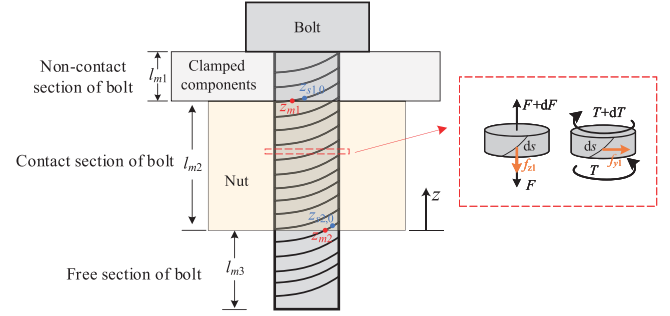


Fig. 4. Element force analysis.

The tensile and torsional equilibrium equations of the bolt can be obtained from the element force analysis.

$$dF + f_{z1} ds = 0, \quad dT - f_{y1} a ds = 0.$$

Combining the coordinate transformation equations and the relationship between stress and strain,

$$F = E_s A_s \frac{du}{dz_s}, \quad T = G_s I_p \frac{d\varphi}{dz_s}, \quad ds = \frac{dz}{\sin \alpha},$$

the above equilibrium equations can be expressed as

$$\begin{aligned}E_s A_s \frac{d^2 u}{dz_s^2} + \left(\frac{\mu}{\cos \beta} - \cot \alpha \right) k_s \delta_s &= 0, \\ G_s I_p \frac{d^2 \varphi}{dz_s^2} - \left(\mu \frac{\cot \alpha}{\cos \beta} + 1 \right) a k_s \delta_s &= 0,\end{aligned}$$

where A_s denotes the bolt cross sectional area, I_p denotes the second polar moment of area of the bolt cross-section, G_s is shear modulus of the bolt. u and φ represent the displacement and rotation angle of the bolt, respectively.

3.3. Supplementary deformation conditions

In Shikun Lu's study, it can be seen that the stress in the nut is very small compared to the stress in the bolt [12]. Therefore, the overall compressive deformation of the nut can be neglected when only the stress distribution in the bolt is considered. In this paper, the deformation relationship between the thread contact points of bolts and nuts is investigated by considering the deformation of the bolt and nut threads as well as the overall deformation of the bolt.

In the process of bolt tightening, the contact points of the bolt and nut satisfy the following deformation relationship:

i. Matching contact points occupy the same position along the z -axis. Assume that the stiffness of the clamped components is k_c and the compressive deformation of the clamped components is δ_c .

$$z_s + u - \delta_s \cos \alpha - \delta_c = z_m + \delta_m \cos \alpha. \quad (5)$$

ii. Assuming no deformation of the reference helix, the corresponding angles of the matching contact points are the same.

$$\theta_m(z_m) = \theta_s = \theta_{s0}(z_s) - \varphi(z_s), \quad (6)$$

where

$$\theta_m = \frac{2\pi z_m}{p}, \quad \theta_{s0} = \frac{2\pi z_s}{p}.$$

Substituting Eqs. (5)–(6) into the equilibrium equations, we can obtain that

$$E_s A_s \frac{d^2 u}{dz_s^2} + \left(\frac{\mu}{\cos \beta} - \cot \alpha \right) \frac{k_s k_m}{(k_s + k_m) \cos \alpha} \left(u + \frac{p\varphi}{2\pi} - \delta_c \right) = 0, \quad (7)$$

$$G_s I_p \frac{d^2 \varphi}{dz_s^2} - \left(\mu \frac{\cot \alpha}{\cos \beta} + 1 \right) \frac{ak_s k_m}{(k_s + k_m) \cos \alpha} \left(u + \frac{p\varphi}{2\pi} - \delta_c \right) = 0. \quad (8)$$

3.4. Boundary conditions

Boundary conditions for bolts vary in different installation environments. In the assembly process studied in this paper, it is assumed that the position of the nut is fixed and the bolt is subjected to tightening torque. In this case, the torque and axial force acting on the bolt do not exist at the free end of the nut. At the loaded face of the nut, the torque acting on the bolt is the thread contact torque T_b , and the displacement of the bolt can be calculated from the tensile deformation of the bolt located in the clamped components. The specific expression form is shown in Eqs. (9)–(10).

At the boundary $z_m = 0$, the tension and torque are both zero.

$$E_s A_s \frac{du}{dz_s} \Big|_{z_m=0} = 0, \quad G_s I_p \frac{d\varphi}{dz_s} \Big|_{z_m=0} = 0. \quad (9)$$

At the boundary $z_m = l_{m2}$, the displacement and torque satisfy

$$u \Big|_{z_m=l_{m2}} = -\frac{Fl_{m1}}{E_s A_s} - \delta_c, \quad G_s I_p \frac{d\varphi}{dz_s} \Big|_{z_m=l_{m2}} = T_b. \quad (10)$$

By using the boundary conditions Eqs. (9)–(10), the equilibrium equations in Eqs. (7)–(8) can be solved and it is possible to obtain information about all displacements and forces on the bolt during the tightening torque addition process.

4. Stress redistribution in the bolt after torque removal

After removing the torque from the bolt head, the friction moment on the contact surface of the bolt head will change to achieve equilibrium again. Besides, the direction of the friction between the threads may also change. Therefore, the thread contact stress may be redistributed. Different stress distribution cases are analyzed below.

4.1. Force analysis

During torque addition, the tightening torque is equal to the sum of the underhead friction torque T_f and the thread contact torque T_b . After torque removal, assuming that the maximum static friction on the contact surfaces is equal to the sliding friction, whether the thread contact stress changes or not is related to the magnitude of T_f and T_b .

If $T_f > T_b$, the frictional moment on the surface of bolt head is sufficient to counteract the thread contact torque, so the stress distribution in the bolt will not change. At this time, the magnitude of the underhead friction moment is equal to T_b , and the direction is opposite to the thread contact torque.

If $T_f < T_b$, the maximum static friction moment on the surface of bolt head cannot counteract the thread contact moment T_b , so the friction distribution on the thread contact surfaces changes and a new equilibrium is achieved. Suppose the friction on the thread close to the bolt head is in the opposite direction, with the quasi-sliding state still satisfied. The thread stress and deformation away from the bolt head remain unchanged. Therefore, the force of the bolt can be divided into

two parts according to the direction of friction, as shown in Fig. 5: the stress invariant segment of $[0, z_1]$, and the stress redistribution segment of $[z_1, l_{m2}]$. The dividing point between the two segments is $z = z_1$. The resultant force and torque of the two segments are F_1, T_{b1} and F_2, T_{b2} , respectively.

In $[0, z_1]$ segment, since the stress does not change, the relationship between T_{b1} and F_1 satisfies Eq. (2). In $[z_1, l_{m2}]$ segment, since the friction on the thread is in the opposite direction, according to the same method of derivation as in Eq. (2), one can obtain that

$$T_{b2} = F_2 a \tan(\alpha - \xi). \quad (11)$$

The directions of T_{b2} depends on the magnitude of α and ξ . If α is greater than ξ , the direction of T_{b2} does not change, i.e., it is in the same direction as T_b . If α is less than ξ , the direction of T_{b2} is opposite to the direction of T_b , as shown in Fig. 5. Different types of stress distribution and their conditions are detailed in Fig. 6. In fact, these conditions are related to the magnitude of the underhead coefficient of friction μ_t and the coefficient of friction between the threads μ .

According to the equilibrium equation of torque, one can obtain

$$T_{r1} = F_2 a \tan(\alpha - \xi) + F_1 a \tan(\alpha + \xi), \quad (12)$$

where T_{r1} is the underhead friction torque after stress redistribution. T_{r1} also satisfies the Coulomb's friction law

$$T_{r1} = \mu_t (F_1 + F_2) r_f. \quad (13)$$

Therefore, the relationship between F_1 and F_2 can be obtained through Eqs. (12)–(13).

$$F_2 = \frac{a \tan(\alpha + \xi) - \mu_t r_f}{\mu_t r_f - a \tan(\alpha - \xi)} F_1. \quad (14)$$

4.2. Tensile and torsional equilibrium equations

The torque and axial force distribution along the bolt are denoted by $M_i(z)$ and $N_i(z)$, respectively. The subscript $i = 1$ represents $[0, z_1]$ segment, and $i = 2$ represents $[z_1, l_{m2}]$ segment. $F(z)$ and $T(z)$ are the axial force and torque distribution during the tightening torque addition process in Section 3.

In $[0, z_1]$ segment, since the stress does not change, the torque and axial force distribution satisfy $M_1(z) = T(z)$ and $N_1(z) = F(z)$.

In $[z_1, l_{m2}]$ segment, due to the change in direction of friction between the threads, the thread contact force can be expressed in e_3 coordinate system as $(0, -\mu f, -f)$. Assuming that the supplementary deformation conditions in 3.3 are still satisfied, the tensile and torsional equations can be established with the same method in Section 3.

$$E_s A_s \frac{d^2 u}{dz_s^2} - \left(\frac{\mu}{\cos \beta} + \cot \alpha \right) \frac{k_s k_m}{(k_s + k_m) \cos \alpha} \left(u + \frac{p\varphi}{2\pi} - \delta_c \right) = 0, \quad (15)$$

$$G_s I_p \frac{d^2 \varphi}{dz_s^2} + \left(\mu \frac{\cot \alpha}{\cos \beta} + 1 \right) \frac{ak_s k_m}{(k_s + k_m) \cos \alpha} \left(u + \frac{p\varphi}{2\pi} - \delta_c \right) = 0. \quad (16)$$

4.3. Boundary conditions in stress redistribution segment

After stress redistribution, the stress distribution in the section $[0, l_{m1}]$ is the same as that before the torque removal. Therefore, at the boundary $z_m = z_1$, the magnitude of torque and axial force has not changed according to the continuity condition. At the boundary $z_m = l_{m2}$, the torque acting on the bolt is equal to the underhead friction torque T_{r1} , and the displacement of the bolt can be calculated from the tensile deformation of the bolt located in the clamped components. The specific expression form is shown in Eqs. (17)–(18).

At the boundary $z_m = z_1$, according to the continuity condition, The torque and axial force satisfy

$$M_2(z_1) = M_1(z_1) = T(z_1), \quad N_2(z_1) = N_1(z_1) = F(z_1), \quad (17)$$

where T and F are the torque and axial force distribution during the torque addition process in Section 3.

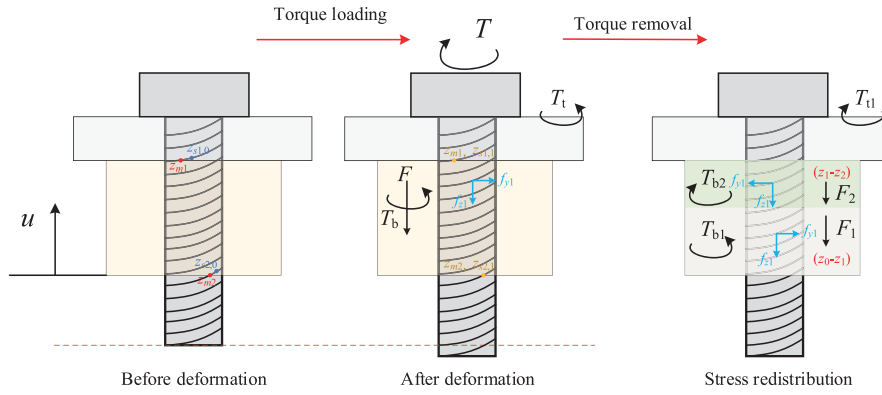


Fig. 5. Stress redistribution analysis.

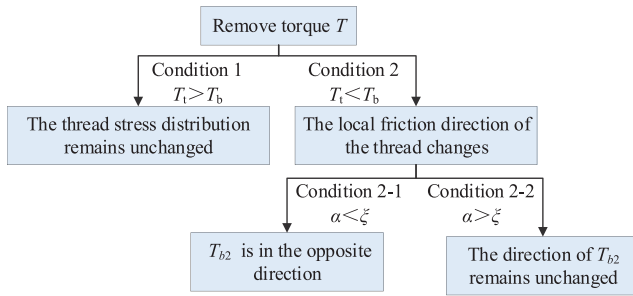


Fig. 6. Different types of stress distribution analysis.

At the boundary $z_m = l_{m2}$, the displacement and torque satisfy

$$u(l_{m2}) = -\frac{(F_1 + F_2)l_{m1}}{EA} - \delta_c, \quad M_2(l_{m2}) = T_{r1}. \quad (18)$$

Since the position of the dividing point $z = z_1$ is unknown, additional condition is needed. Here, the rotation angle at point $z = z_1$ is considered to be the same as that during torque addition. Combined with the above boundary conditions, differential equations in Eqs. (15)–(16) can be solved.

5. Numerical simulation analysis

Through the developed static analytical model, comprehensive displacement fields and stress distribution characteristics within the bolted joint during the entire assembly process can be quantitatively determined. To further investigate the mechanical behavior predicted by the proposed model, a systematic numerical simulation study was conducted, focusing on two critical aspects: Firstly, as demonstrated in Section 4, the stress distribution exhibits significant variations upon removal of the tightening torque, necessitating detailed analysis of the transition conditions between different stress patterns. Secondly, the deformation distribution along the bolt axis was thoroughly examined through torque-displacement relationship analysis.

The numerical simulations were performed using the geometric and material parameters specified in Table 1. To ensure the representativeness of the simulation results, two distinct bolt configurations were implemented: M10 bolts with round heads and M20 bolts with hexagonal heads, thereby covering typical bolt geometries in engineering applications.

5.1. Parameter research of stress redistribution phenomenon

When the geometry and material of the bolt are determined, the distribution of internal forces in the bolt after removal of the tightening torque is mainly determined by the coefficient of friction on the contact

Table 1

Specific parameters of the bolt.

Parameters	M10 bolt	M20 bolt	Unit
Bolt head cross-section diameter	16	30	mm
Thread pitch, p	1.5	2.5	mm
Thread effective diameter, d_p	9	18.4	mm
Thread lead angle, α	$\pi/60$	$\pi/60$	rad
Half of thread included angle, β	$\pi/6$	$\pi/6$	rad
Young's modulus, E	206	206	GPa
Poisson's ratio, ν	0.3	0.3	–

surface. In the following, the distribution of torque and axial force for different friction coefficients μ_t and μ is investigated. ISO metric M20 bolts shown in Table 1 are used as the subject of this study. The stiffness of the clamped components $k_c = 1 \times 10^{11}$ N/m.

The conditions of different types of stress distribution are related to the magnitude of the underhead coefficient of friction μ_t and the coefficient of friction between the threads μ . The relationship between the two friction coefficients and these conditions is illustrated in Fig. 7. As shown in the blue area, Condition 1 occurs when the underhead coefficient of friction μ_t is relatively large. The distribution of torque and axial force in the bolt remains unchanged compared to the distribution during the torque addition process. Condition 2-1 occurs when the coefficient of friction between the threads μ is relatively large, as shown in the gray area. The direction of the thread friction torque T_{b2} will be reversed in this condition. Condition 2-2 occurs only when both friction coefficients are small, as depicted by the red area in Fig. 7. When the friction coefficient μ_t satisfies a specific interval around 0.06, with the change of μ , all three types of distribution occur. When μ_t is greater than a specific value around 0.9, no matter how μ changes, the stress distribution in the bolt is always constant after removal of the applied torque T .

The axial force and torque distribution curves at different friction coefficients are depicted in Fig. 8. The length of the clamped components $l_{m1} = 20$ mm, the length of thread engagement $l_{m2} = 50$ mm, the length of the screw is $l = 90$ mm, and the outer diameter of the nut is 30 mm. During general assembly process, the specified preload requirements need to be met. Here the axial preload is supposed to be 3 kN. When $\mu = 0.1$, as shown in Fig. 8(a), $\tan(\alpha - \xi)$ is always less than zero, which means Condition 2-2 will not occur as μ_t changes. Therefore, the torque distribution does not change (Condition 1, gray lines, $\mu_t = 0.2$) or the direction of the torque is reversed in the part close to the loaded end (Condition 2-1, red and blue lines, $\mu_t = 0.1$ and 0.05). $\tan(\alpha - \xi)$ is always greater than zero when $\mu = 0.03$ as shown in Fig. 8(b), so that Condition 2-1 cannot be achieved. The direction of the torque on the bolt is always the same. All three types of distribution occur as μ changes when $\mu_t = 0.05$, as illustrated in Fig. 8(c). Besides, it is worth noting that the axial force distribution remains essentially unchanged after removal of the tightening torque.

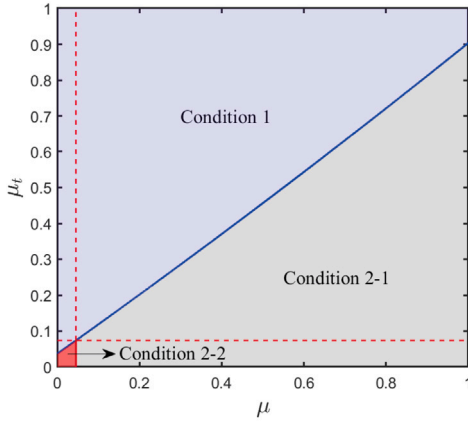


Fig. 7. Parameter analysis for different conditions.

5.2. Load and deformation analysis along the bolt

According to the stress distribution model during and after the torque addition process, the load and deformation information of the whole bolt can be obtained from the tightening torque applied to the bolt. In order to perform simple calculation, assuming an applied torque of 10 N m for the M10 bolt shown in Table 1, the coefficient of friction $\mu_t = 0.1$ and $\mu = 0.2$. The preload force can be calculated as 5.17 kN and the thread contact torque is 6.69 N m. The length of the clamped components $l_{m1} = 20$ mm, the length of thread engagement $l_{m2} = 30$ mm, the length of the screw $l = 60$ mm, and the outer diameter of the nut is 16 mm. The stiffness of the clamped components $k_c = 1 \times 10^8 / 5 \times 10^8 / 1 \times 10^9$ N/m. Here the coordinate z_m indicates the distance of a point on the bolt from the bottom of the bolt. Then the load and deformation distribution along the whole bolt during the torque addition process are depicted in Fig. 9. The bolt is not subjected to any external force in the free section, while in the non-contacting section it is subjected to a constant axial preload force P and thread contact torque T_b . And the relative axial displacement $\Delta u(l_{m3})$ and relative rotation angle $\Delta \varphi(l_{m3})$ of the bolt in the non-contact section can be obtained by Eq. (19).

$$\Delta u(l_{m3}) = \frac{Pl_{m3}}{E_s A_s}, \quad \Delta \varphi(l_{m3}) = \frac{T_b l_{m3}}{G_s I_p}. \quad (19)$$

The distribution of forces and displacements is similar for the case with different stiffness coefficients k_c . For example, when $k_c = 1 \times 10^8$ N/m, the bolt is not subjected to any external force in the free section, so the torque and axial force are zero, as shown in Fig. 9(a, b). Simultaneously, the rotation angle (Fig. 9(c)) and displacement (Fig. 9(d)) of the bolt are constants. In the contact section, due to the complex force and contact deformation between the threads, the load and deformation of the bolt show a non-linear variation. In the non-contact section, the threads of the bolt do not bear any external force, and the deformation of the bolt can be regarded as a simple combination of linear tensile and torsional deformation, with the angle and displacement showing a linear distribution along the bolt. Here the rotation angle in the free section can be considered as the rigid rotation angle of the bolt, and the rotation angle in the other two sections can be considered as the sum of deformed and rigid rotation angle of the bolt. Therefore, during bolt assembly, the rigid rotation angle of the bolt is much larger than the rotation angle due to deformation. In addition, as the stiffness coefficient k_c changes, the torque and axial force distribution of the bolt remains consistent, while the rotation angle and displacement distribution changes considerably. This is because the change in stiffness coefficient affects the magnitude of the initial displacement of the boundary and has no effect on the force distribution. As shown in Fig. 9(c, d), the lower the stiffness coefficient k_c , the higher the rotation

angle and axial displacement of the bolt. It is worth noting that the axial displacements of the bolt with different stiffness coefficient k_c are not the same close to the bolt head boundary, as shown in the enlarged view in Fig. 9(d). This is because the displacement at the boundary is related to the stiffness of the clamped components. The greater the stiffness of the clamped components, the smaller the displacement at the boundary.

Therefore, based on the tightening torque applied to the bolt, we can obtain information about the angle of the bolt head and the overall displacement of the free section of the bolt. This provides theoretical support for the use of corner control to add bolt preload in engineering design.

After the previous discussion, the stress distribution in the bolt may change after removing the tightening torque. Therefore, in the following discussion, the change in stress distribution after removing the tightening torque is investigated based on the previous conditions in this section. The stiffness of the clamped components $k_c = 1 \times 10^8$ N/m. The comparison of the load and displacement of the bolt in these two cases is shown in Fig. 10.

After removing the tightening torque, the resultant thread contact torque becomes smaller, satisfying Condition 2-1 in Section 5.1, as depicted in Fig. 10(a). The torque in the contact section first increases gradually and is the same as the torque distribution during the tightening torque addition process. Then the torque decreases after reaching the critical point on the bolt. The critical point here is approximately 48 mm from the bottom of the bolt. Ultimately, the maximum torque on the bolt is reduced to 3.3 N m. The distribution of axial forces is essentially the same in both cases, except that the axial forces after torque removal are slightly lower than those before torque removal, as depicted in Fig. 10(b). It can be seen that after removing the tightening torque, the preload force of the bolt decreases slightly by roughly 18 N. Therefore, this model can explain the decrease in preload after bolt assembly. As shown in Fig. 10(c), slight reduction occurs in rotation angle of the bolt after removal of tightening torque. As illustrated in the enlarged view of the figure, the maximum rotation angle of the bolt is reduced by less than 0.01° compared to that before torque removal. The axial displacement also changes slightly, but the reduction is extremely small, as shown in Fig. 10(d).

Condition 2-2 will occur if the two friction coefficients change such that $\mu_t = 0.03$ and $\mu = 0.04$. Here the stiffness of the clamped components $k_c = 1 \times 10^9$ N/m. The tightening torque is still 10 N m. The preload force is 15.66 kN and the thread contact torque is 6.98 N m in this case. Comparing to the previous case, it can be seen that the smaller the friction coefficient, the more preload can be applied at the same tightening torque. The load and deformation distribution of the bolt in this case is depicted in Fig. 11.

In the contact section, the torque always maintains an upward trend, as shown in Fig. 11(a). Before the critical point $z_m = 45.3$ mm, the torque distribution is the same as that during the tightening torque addition process. Then the torque increase slows down after reaching the critical point. Ultimately, the maximum torque on the bolt is reduced to 3 N m. Similar to the previous case, the distribution of the axial force in the bolt remains essentially unchanged, but the final preload is slightly reduced by about 60 N, as shown in Fig. 11(b). Besides, slight reduction occurs in rotation angle and axial displacement of the bolt after removal of tightening torque can also be observed in Fig. 11(c, d).

5.3. Comparison with existing literature for load distribution

Yamamoto developed a load distribution model for bolts subjected to unidirectional tension [10]. Our team also conducted experimental measurements of the stress distribution of bolts after assembly [24]. In order to verify the validity of the theoretical model proposed in this paper, the load distribution obtained from the theoretical model is compared with the results of the above studies in this section. Stress

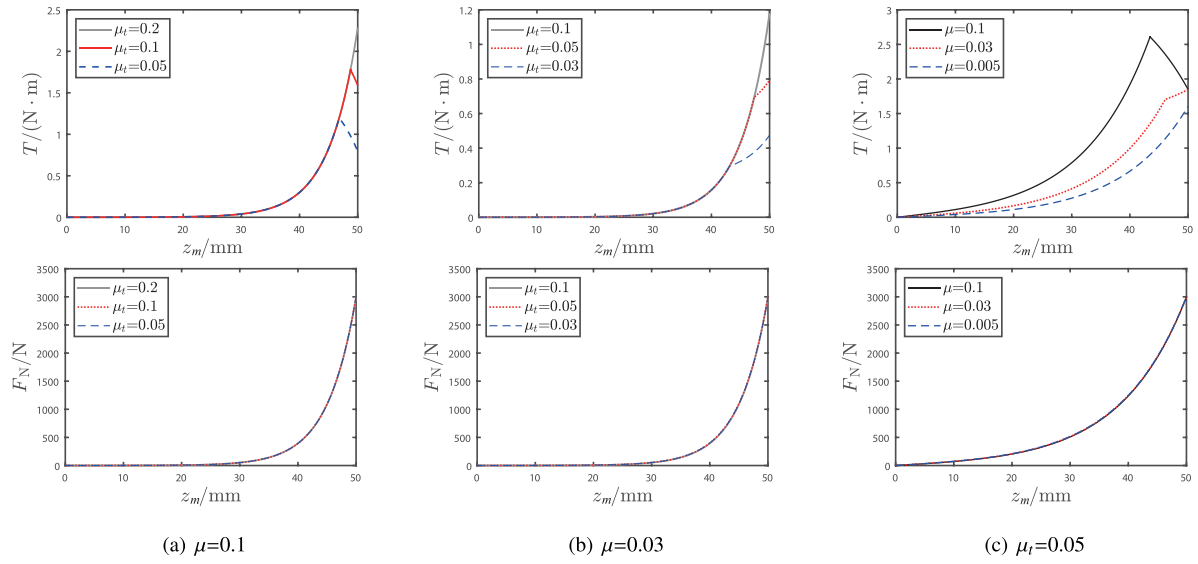


Fig. 8. Axial force and torque distribution in the thread contact section.

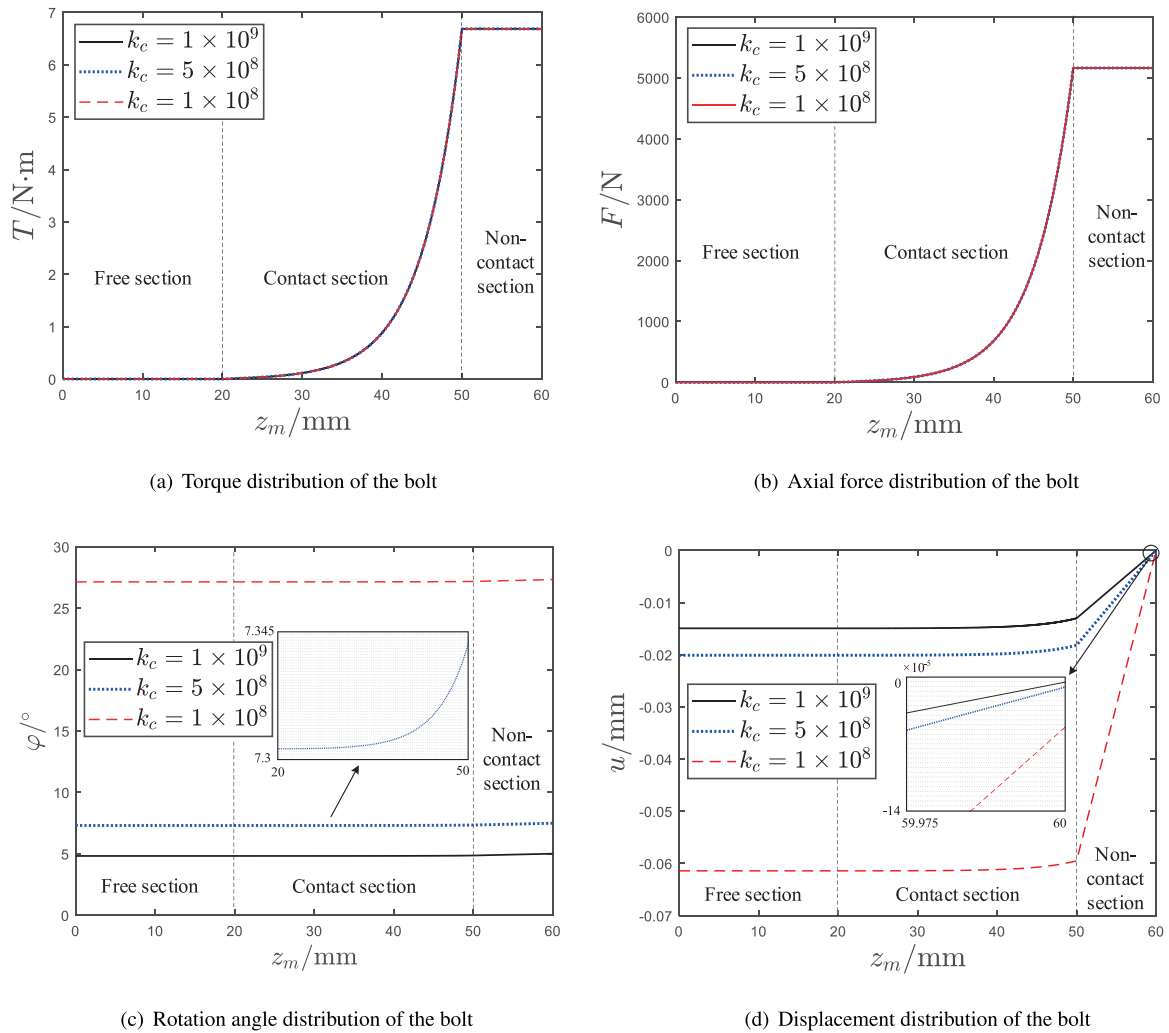


Fig. 9. Load and deformation distribution during the torque addition process.

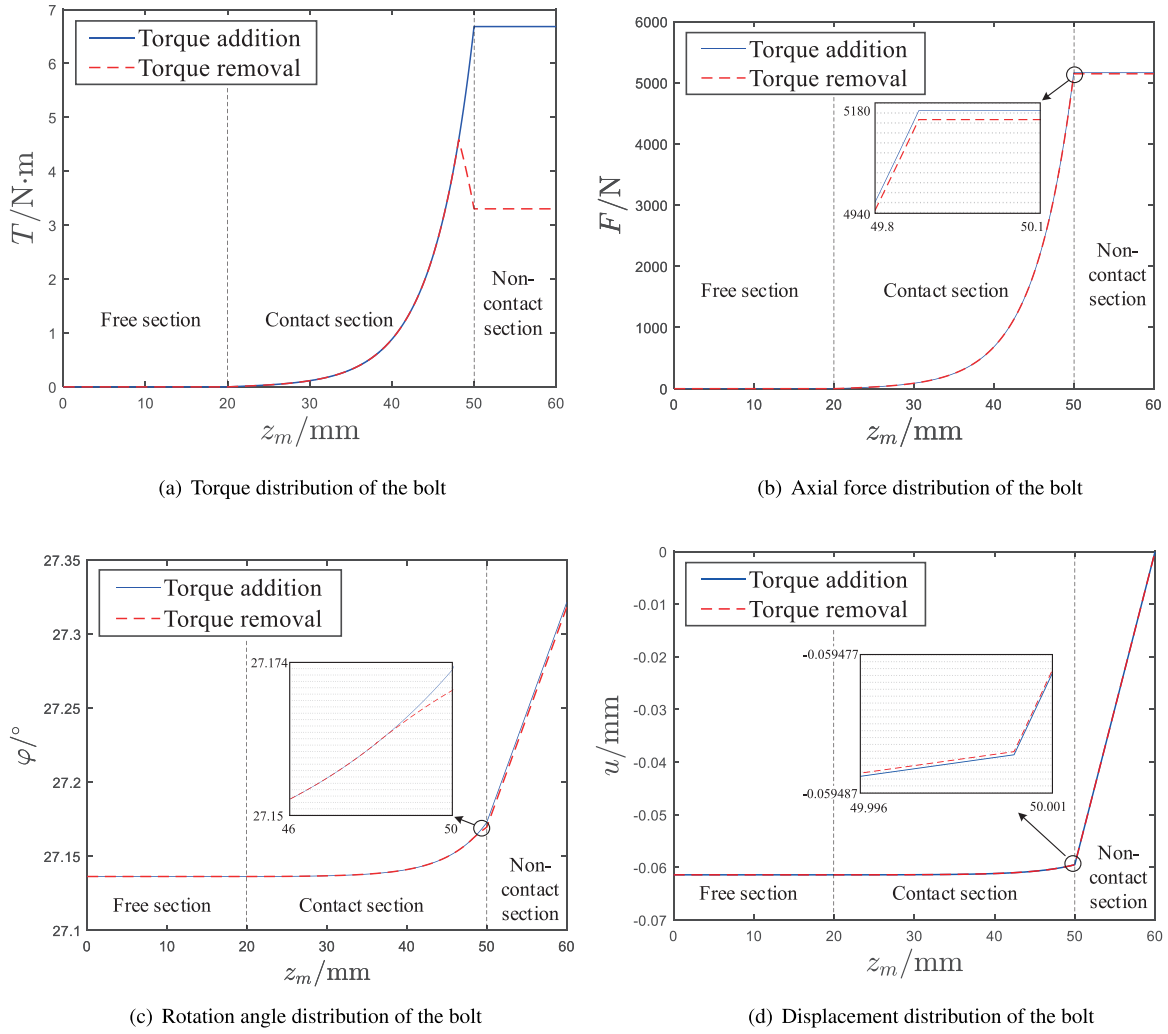


Fig. 10. Load and deformation distribution.

distribution analyses are carried out for the M20 bolts listed in Table 1. The bolt axial tensile stress σ can be calculated by Eq. (20).

$$\sigma = F/E_s, \quad (20)$$

where F is the axial force. The tightening torque T is 3/4 / 5/6 N m, and the length of thread engagement $l_{m2} = 48$ mm. The friction coefficients $\mu_t = 0.06$ and $\mu = 0.06$. z_s indicates the axial coordinate in the contact section of the bolt. The stress distribution is depicted in Fig. 12.

As can be observed from Fig. 12, the theoretical model proposed in this paper is basically consistent with the Yamamoto model and the experimental data obtained from OFDR experimental measurements under different tightening torques. The axial stress obtained by the proposed model is slightly higher than that obtained by the Yamamoto model, with a better overlap with the experimental data. Besides, in OFDR experiments, the phenomenon of decreasing stress is found. Whereas, the proposed model has a gradual increase in stress as in the Yamamoto model.

6. Experimental verification

In this section, experiments are conducted on the relationship between tightening torque and preload and the relationship between tightening torque and bolt head rotation angle. The experimental results are analyzed and compared with the proposed theoretical model. M10 bolts are used in experiments, as shown in Table 1. In order to

facilitate the installation, the bolt head is hexagonal with a length of 16 mm on opposite sides. The hexagonal nuts used in experiments are 17 mm on opposite sides.

The experiment setup is illustrated in Fig. 13. It consists of three main parts: the bolt installation platform, the pressure data acquisition system and the torque/angle measuring instrument. The bolt installation platform is responsible for the assembly of the bolt, including test bench, pressure sensor, bolt, nut, thrust bearing and other components. The nut is fixed at the designated position inside the test bench. The height of the nuts used in this experiment is 10 mm and 30 mm. The pressure sensor is SA-YB-HQ/1T produced by Shiao Technology Co., Ltd. with a measuring error of less than 0.5%. Through the use of the thrust bearing, the influence of friction between the bolt head and the contact surface on the preload can be eliminated. In this experiment, when it is necessary to study the underhead friction coefficient, the thrust bearing will be replaced by a washer. The pressure data acquisition system is used to measure the preload of the bolt. The data acquisition system used in this system is SA1808B2 produced by Shiao Technology Co., Ltd. The accuracy of the system is better than 0.5%. The torque/angle measuring instrument used in the experiment is the digital angle torque wrench SDA-030-22 manufactured by the Stanley Works (Shanghai) Co., Ltd. This digital angle torque wrench is used to measure the tightening torque with an error of no more than 0.17% and the rotation angle of the bolt head with an error of no more than 2°.

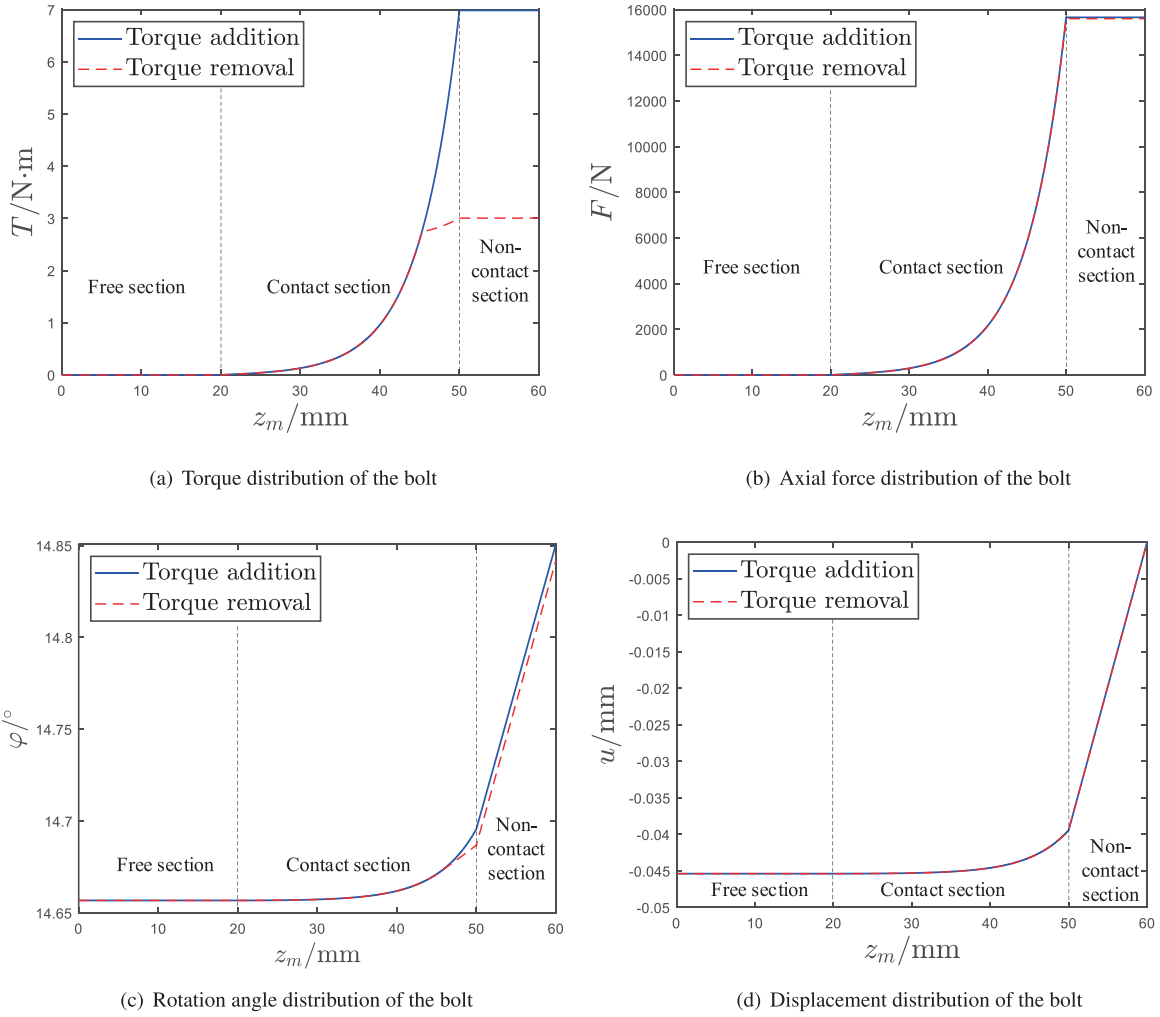


Fig. 11. Load and deformation distribution.

During installation, the test bench is fixed on the experiment table using clamps. The pressure sensor is positioned on the contact surface of the test bench to measure the preload. The bolt is tightened through the pressure sensor and the washer or the thrust bearing onto the bench using the torque wrench. The tightening torque and the rotation angle of the bolt head can be measured by the torque wrench. The pressure sensor are then connected to the pressure data acquisition system. The corresponding preload can be measured at the applied torque.

In the experiment, the rotation angle of the bolt head is controlled by means of the torque wrench, and the values of torque and preload at this point are recorded. The rotation angle of the bolt head is set to be 5°, 7°, 10°, 12°, 15°, 17°, and 20°. This experiment is divided into two groups: one using a thrust bearing and the other using a washer instead of the thrust bearing. Both 10 mm and 30 mm height nuts are used in each group of tests. Each set of tests is repeated three times to obtain the average experimental data.

The torque-preload data are shown as blue and red dots in Fig. 14. The black line represents the theoretical linear relationship between tightening torque and preload force, as shown in Eq. (3). This linear relationship can be obtained by fitting the experimental data. From Fig. 14, it can be seen that the positive proportionality between preload and torque is satisfied in different groups of tests. The inclusion or exclusion of thrust bearings in the experiment determines the magnitude of the proportionality coefficient k_{FT} . The proportionality coefficient k_{FT} without the thrust bearing is significantly higher than that with the thrust bearing. In fact, in the tests with thrust bearings, the tightening

torque is equal to the thread contact torque because the underhead friction coefficient is eliminated, so the relationship between the tightening torque and the preload force satisfies Eq. (2), leading to the reduction of k_{FT} .

Combining Eqs. (2) and (3) and the experimental curves, the friction coefficient identification can be carried out. Based on Eq. (2) and the proportionality coefficient k_{FT} with the thrust bearing, the thread friction coefficient μ can be calculated. Combined with Eq. (3) and proportionality coefficient k_{FT} with the thrust bearing, the underhead friction coefficient μ_t can then be calculated. For the case of 10 mm height nut, $\mu = 0.147$, and $\mu_t = 0.087$. For the case of 30 mm height nut, $\mu = 0.135$, and $\mu_t = 0.084$. The consistent underhead friction coefficient μ_t calculated using two different heights of nuts demonstrates the validity of the experimental results and verifies the accuracy of the theoretical results.

The relationship between the rotation angle of the bolt head and the tightening torque is shown as the blue and red dots in Fig. 15. From the experimental data, it can be seen that the rotation angle and tightening torque basically satisfy the linear relation. As the torque increases, the rotation angle of the bolt head increases. At the same torque, the bolt head with a nut height of 30 mm has a greater rotation angle than the bolt head with a nut height of 10 mm.

In order to verify the correctness of the model, we substitute the friction coefficients calculated above into the theoretical model and choose the stiffness of the clamped components as 7×10^7 N/m, and obtain the theoretical results of the bolt head rotation angle, as shown by the red and blue curves in Fig. 15. Theoretical results indicate that the

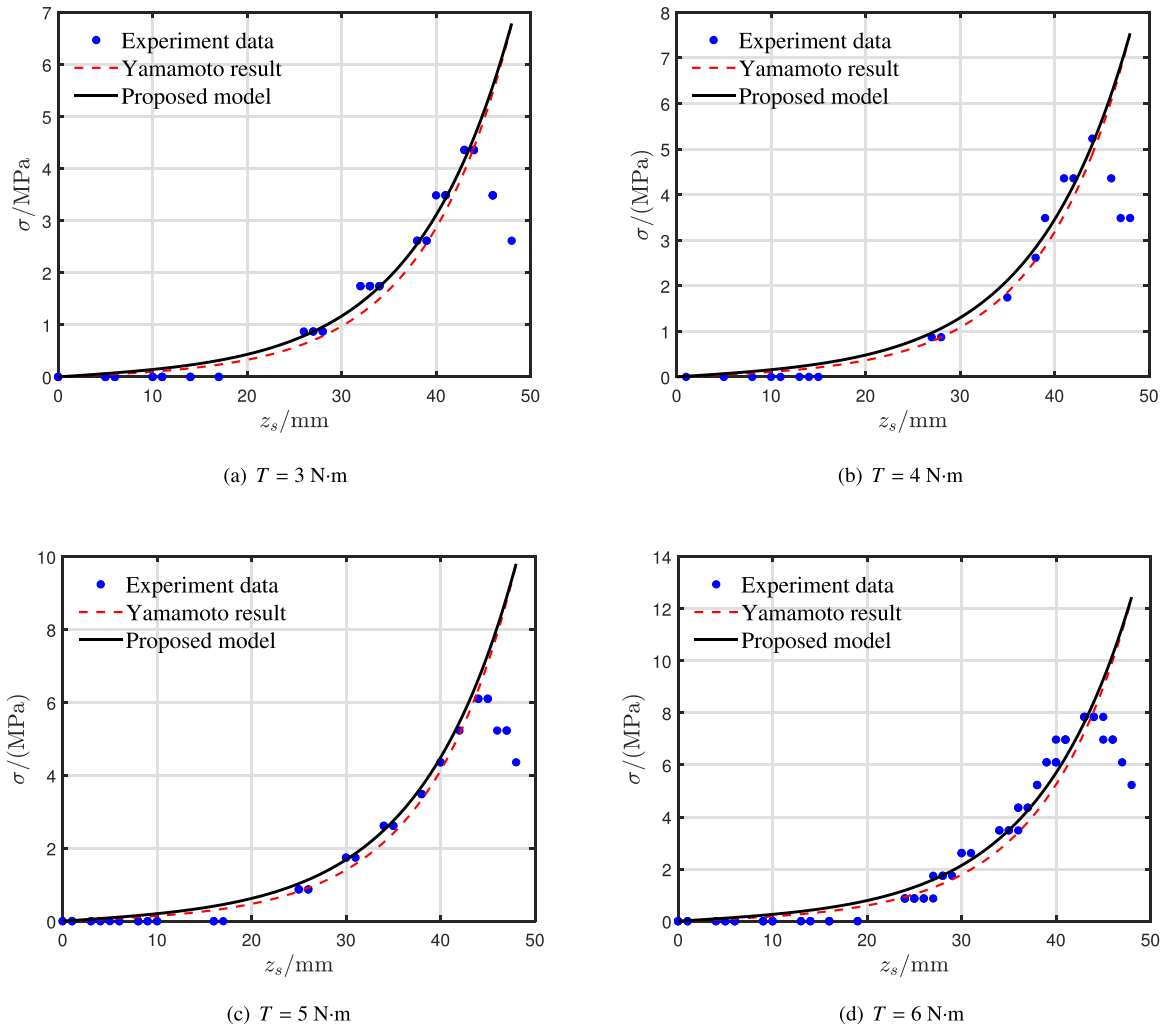


Fig. 12. Stress distribution in the bolt.

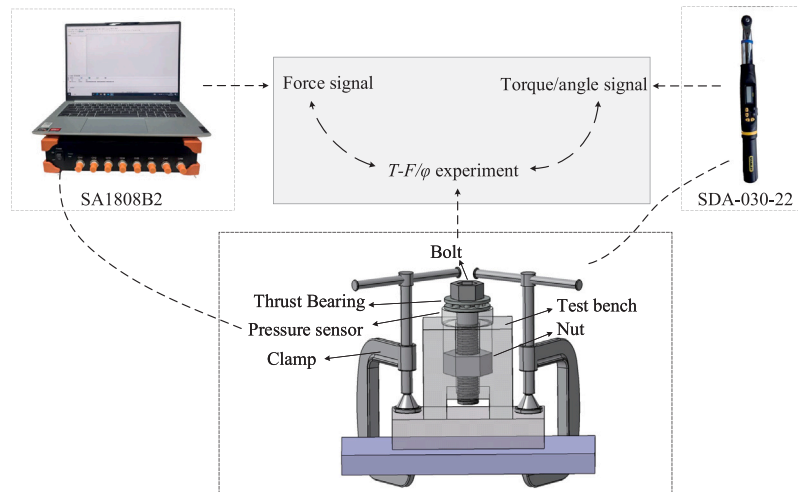


Fig. 13. Experiment setup.

relationship between rotation angle of bolt head and tightening torque is basically linear, and the theoretical curves are in great agreement

with the experimental data. The maximum error between experimental and theoretical data is 8.6%, as shown in Table 2.

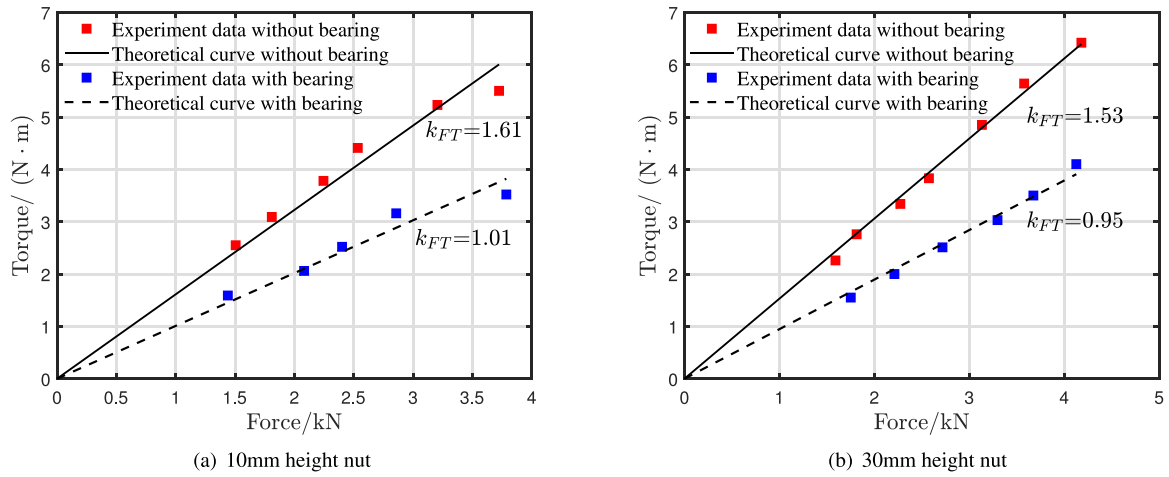


Fig. 14. Torque-preload curves. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 2

The relative error of rotation angle between experimental and theoretical data.

	$l_{m2} = 30 \text{ mm}$					$l_{m2} = 10 \text{ mm}$				
Tightening torque/N m	3.34	3.83	4.85	5.64	6.42	3.78	4.41	5.23	5.5	
Experimental angle/°	10	12	15	17	20	10	12	15	17	
Theoretical angle/°	9.90	11.40	14.40	16.80	19.10	10.66	12.44	14.76	15.53	
Relative error/%	1.0	5.0	4.0	1.2	4.5	6.6	3.7	1.6	8.6	

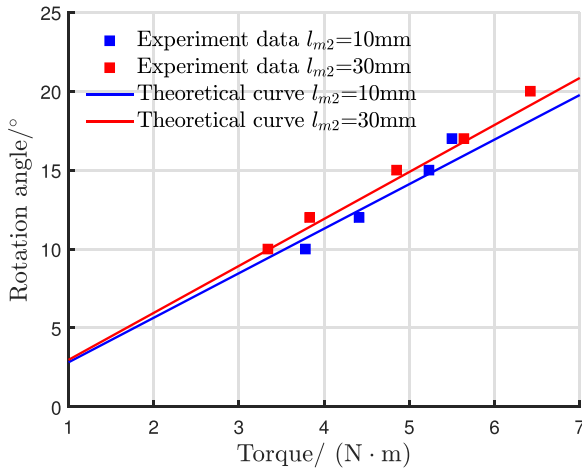


Fig. 15. Torque-rotation angle curves.

7. Conclusions

In this paper, the theoretical modeling and numerical simulation of the axial force and torque distribution during and after bolt assembly are performed. Torque-preload/rotation angle experiments are conducted to validate the proposed model.

First of all, the assembly torque model is established. The linear relationship between tightening torque T and preload force F is formulated. Stress distribution in the bolt during and after assembly process is modeled. The tensile and torsional differential equations are developed by element force analysis of the bolt. Combined with the supplementary deformation conditions and boundary conditions, the force and displacement distribution can be deduced.

Numerical simulation analysis is then carried out. Parametric conditions for the emergence of different types of distribution patterns are analyzed. At different friction coefficients, the stress in bolts after assembly may show different distribution patterns. Based on the tightening torque applied to the bolt, we can obtain information about the

angle of the bolt head and the overall displacement of the free section of the bolt. The effect of the stiffness of the clamped components on the load and displacement distribution is analyzed. The smaller the stiffness of the clamped components, the greater the rotation angle and axial displacement of the bolt. Besides, the proposed model can explain the phenomenon of decreasing preload after assembly. This provides theoretical support for the use of corner control to add bolt preload in engineering design. The comparison with existing literature for load distribution verifies the validity of the proposed model.

Finally, the relationship between tightening torque and preload and the relationship between tightening torque and bolt head rotation angle are obtained through experiments. The linear relationship between tightening torque and preload force can be observed in the tests. Through the use of the thrust bearing, the friction coefficient identification is carried out. Substituting the identified friction coefficients into the theoretical model and selecting the stiffness of clamped components as $7 \times 10^7 \text{ N/m}$, the torque-rotation angle curves are obtained in great agreement with the experimental data, which verifies the validity of the model.

The proposed model contributes to a deep understanding of bolt assembly problems in engineering. This method is a guide for the installation of bolts as well as for safety calibration.

CRediT authorship contribution statement

Zhixiang Li: Investigation, Methodology, Validation, Writing – original draft, Writing – review & editing. **Zhen Zhao:** Conceptualization, Funding acquisition, Methodology, Supervision. **Qingyun Wang:** Methodology, Supervision. **Jiaying Zhang:** Conceptualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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